

AI CODE REVIEWER

Input Code -> AI:
Identifies language
Detects errors
Highlights warnings
Suggests improvements
Rates the code

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Problem Statement:

Many students and beginners face difficulties in identifying errors in their code and understanding proper programming practices. Traditional methods of code review require guidance from experienced programmers, which is not always available. As a result, students struggle to debug their code, improve logic, and learn effectively. Therefore, there is a need for a simple and intelligent system that can automatically analyze code, detect errors, and provide useful suggestions in real-time.



Objectives:

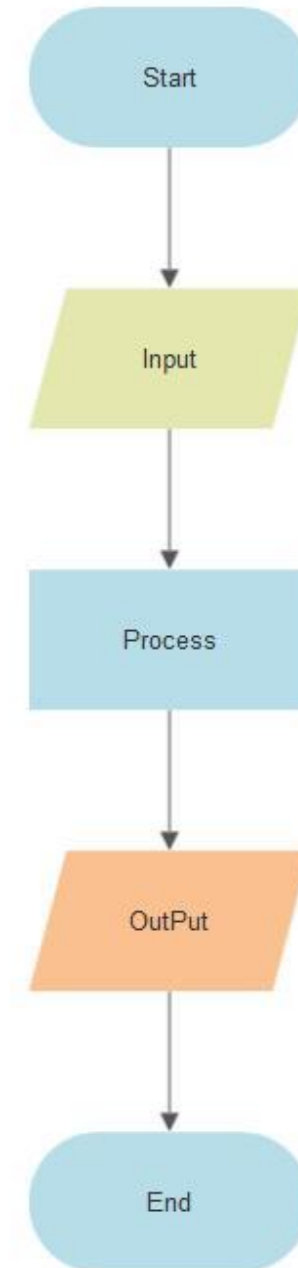
- To develop an AI-based system for analyzing programming code.
- To detect errors and provide accurate feedback to users.
- To help students understand coding mistakes easily.
- To provide useful suggestions for improving code quality.
- To support multiple programming languages.
- To create a simple and user-friendly interface using Streamlit.
- To enable quick and efficient code review without external compilers.

Innovation and Uniqueness:

- Integrates **AI-based code analysis** for intelligent feedback.
- Works **without external compilers**, making it lightweight and easy to use.
- Provides **instant feedback** including errors, warnings, and suggestions.
- Supports **multiple programming languages** in a single platform.
- Offers **different analysis levels** (Quick, Standard, Deep) for flexibility.
- Features a **simple and user-friendly interface** using Streamlit.
- Includes **Dark/Light mode** for better user experience.

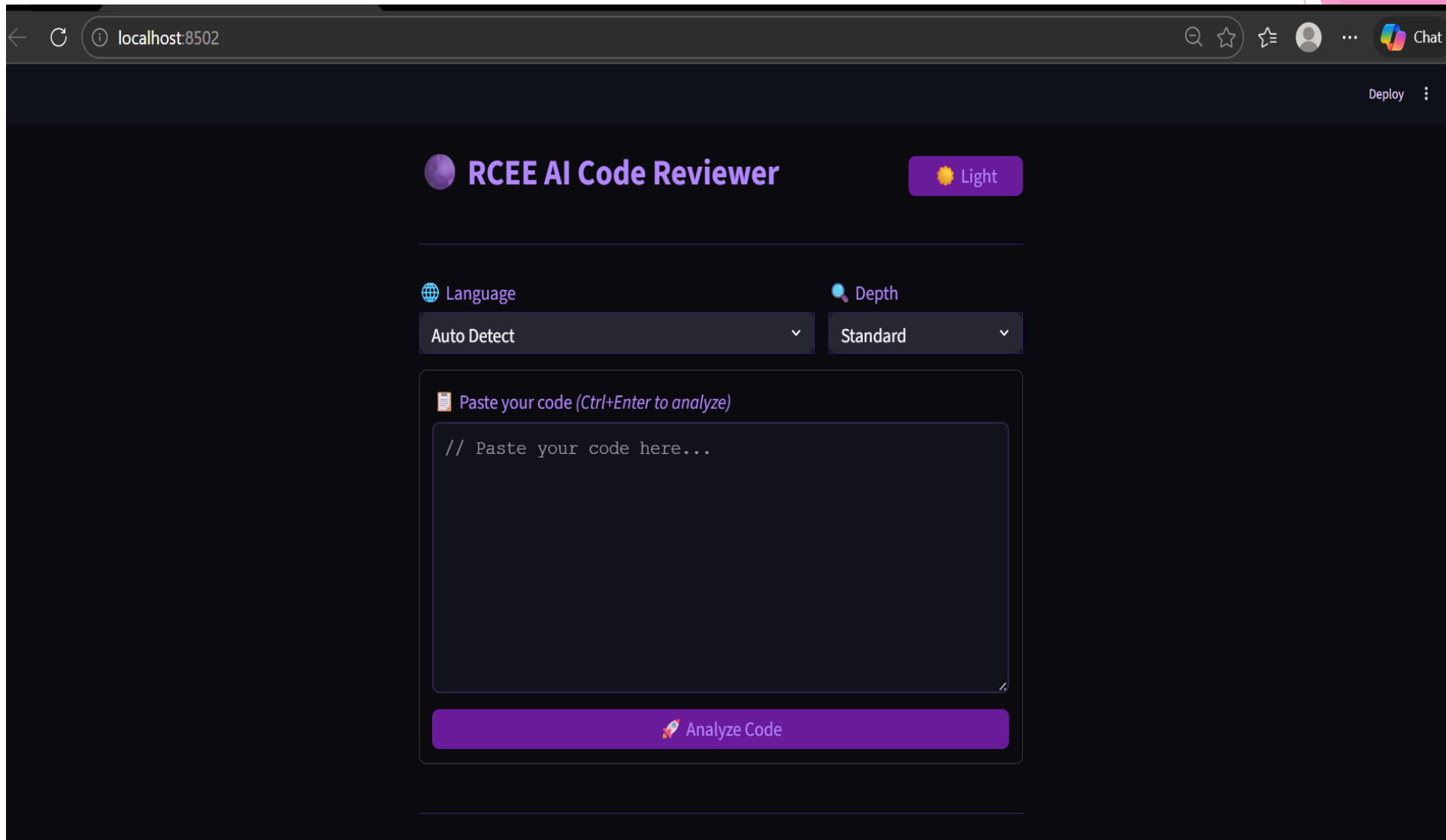
Proposed Methodology:

- User enters code
- Selects language & depth
- Code sent to AI model
- AI analyzes code
- Errors & suggestions generated
- Results displayed



Expected Outcomes:

- The system will successfully **analyze user code**.
- It will **detect syntax errors and issues** in the code.
- It will provide **clear and useful suggestions** for improvement.
- The application will generate an **overall rating for the code quality**.
- Users will be able to **understand mistakes easily**.
- It will help students **improve their coding skills**.
- The system will deliver **quick and efficient results**.



localhost:8502

RCEE AI Code Reviewer

Light

Language

Auto Detect

Auto Detect

Python

Java

C

C++

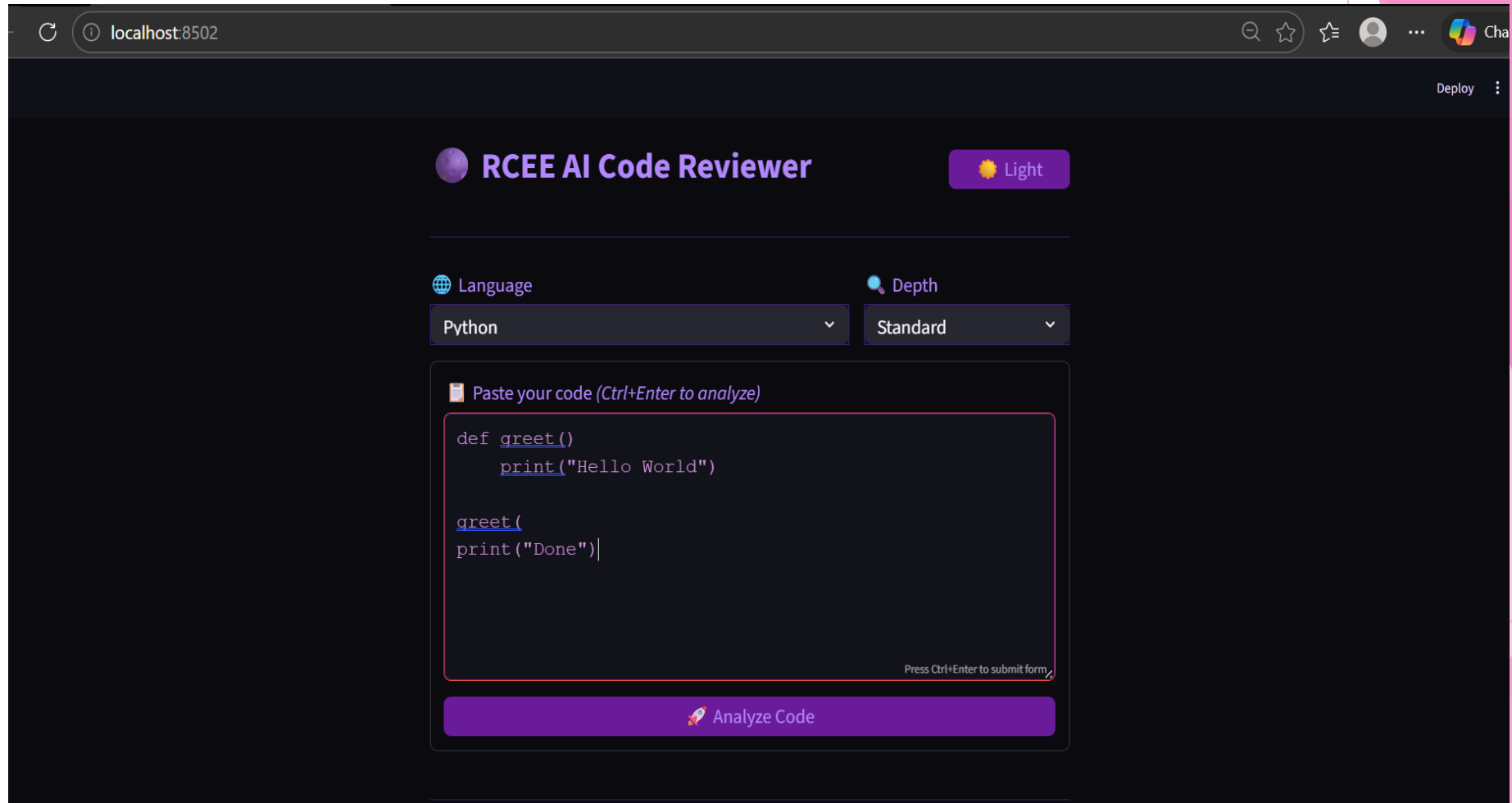
JavaScript

TypeScript

Depth

Standard

Analyze Code



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localhost:8502

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Chat

Deploy ⋮

RCEE AI Code Reviewer

Light

Language

Python

Depth

Standard

Paste your code (Ctrl+Enter to analyze)

```
def greet()  
    print("Hello World")  
  
greet(  
print("Done")
```

Analyze Code

Language: Python

Errors: Line 1: missing colon at the end of function definition, Line 4: missing closing parenthesis

Warnings: ☒ None

Suggestions: add a main guard to ensure greet function is called when the script is run directly

Rating: 2/10

localhost:8502

Chat

RCEE AI Code Reviewer

Light

Language

Python

Depth

Standard

Paste your code (Ctrl+Enter to analyze)

```
def greet():  
    print("Hello World")  
  
greet()  
print("Done")
```

Analyze Code

Language: Python

Errors: ☒ None

Warnings: ☒ None

Suggestions: Consider adding a main function to encapsulate the program entry point

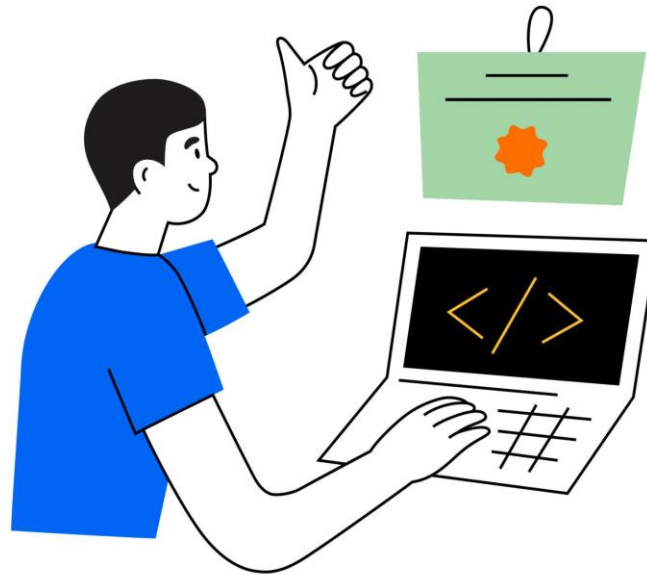
Rating: 8/10

Work Plan / Timeline:

- Week 1:** Requirement analysis and project planning
- Week 2:** Designing the system and UI layout
- Week 3:** Developing the Streamlit interface
- Week 4:** Integrating AI model (Groq API)
- Week 5:** Implementing code analysis features
- Week 6:** Testing and debugging the application
- Week 7:** Final modifications and improvements
- Week 8:** Documentation and PPT preparation

Conclusion:

- The **AI Code Reviewer** is a simple and effective tool for code analysis.
- It helps users **detect errors and improve code quality**.
- The system provides **instant feedback, suggestions, and rating**.
- It is **easy to use** and does not require complex setup.
- The project enhances **learning and coding skills** of students.
- It can be further improved with **advanced AI features** in the future.



The background features abstract, overlapping geometric shapes in various shades of pink and purple, primarily concentrated on the right side of the frame. The shapes include triangles and polygons of different sizes and opacities, creating a layered, modern aesthetic. The text "Thank You" is centered in the white space on the left.

Thank You